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# PLACEMENT DATABASE MANAGEMENT SYSTEM

## ❖ PURPOSE

The purpose of the Placement Database is to define, store, and manage all information involved in campus placements and internships. The system will manage data about students, companies,internships, applications, interviews, offers, joining outcomes, and the people who run the process like, TPO staff and student volunteers.

The main purpose of placement databases is to reduce communications gaps between companies and institutes, reduce manual workload i.e. database makes records easier to analyze while Books or registers are difficult to handle.

- It provides a single source for Placements and internships.
- It enforces fairness like CPI criteria, no backs etc…
- It gives accurate analysis for the institute as per Companies requirements.
- Makes it easier for companies to analyze student’s records and achievements.
- It provides real time placement analytics.
- It provides mentoring support through faculty/alumni guidance.
- It also provides self-analytics dashboards to help students evaluate their progress

# ❖ Intended Audience & Reading Suggestions

### Training and Placement Officers (TPOs)

Use the databases to create placement cycles,approve company drives, verify results, and generate official statistics.

### Student Placement Committee (SPC)

Support TPOs in daily operations such as managing schedules, coordinating students, and ensuring data accuracy.

### Students

Use the platform to create and update profiles, upload resumes, apply to opportunities, and check application status.

### Recruiters/Companies

Post job and internships descriptions, eligibility criteria, and manage applicant pools, assessments, and offer rollouts.

### University Administration & Management

Access placement analytics for performance measurement, accreditation (NAAC/NIRF), and strategic decisions.

### Reading Suggestions

1. scribd.com
2. wikipedia
3. Visit of DAU Placement website

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## ❖ Product Scope

- **Placement cycle organization:** drive creation, slotting, eligibility rules like, CGPA,branch,backlogs,gender etc… application windows, document collection.

- **Process tracking:** Aptitude tests, coding tests, interview rounds, results, offers, accept or reject, joining, upgrades or super-dream conversions.

- **Analytics & reporting:** placement percentage, average/median CTC, role-wise stats, offer to join ratio, diversity metrics.

- **Student - Analytics:** Students can analyze Student's progress like in how many interviews students passed aptitude tests, coding tests and interviews etc.

- **Achievement tracking**: Maintain records of student achievements like awards, competitions, research publications, online certifications, and extracurricular activities. These can be used to:
    - Enhance student profiles for recruiters.
    - Apply achievement-based eligibility filters (e.g., coding competition winners for product companies).

- **Mentoring Support**: Guidance from alumni with mentor feedback to improve placement readiness.

## ❖ Description

The Placement Database Management System is designed to replace the manual, paper-based processes currently used by college placement cells to manage student placements. Existing systems often rely on spreadsheets, emails, and physical records, leading to inefficiencies, errors, and delays in communication. The main goal is to centralize and automate these process. The system will serve as a single point of access for all placement-related activities, including student and company data management, placement scheduling, and notifications.

It will use a centralized database to store student profiles, company details, and placement records, ensuring data consistency and accessibility. The system will support multiple user roles with distinct permissions to ensure secure and role-appropriate access. For example, students can only view and apply to placement opportunities, while Training and Placement Officers (TPOs) have administrative control over data and schedules.

- **Functions**
  - **User Management:** To manage the registrations, deletions, and profile updations.

1. **Administrator Functions:**

   *Account Creation:* The admin can create accounts for students and companies by inputting relevant details like, student information, company information.

   *Account Updates:* Admins can modify user details, such as updating a student's CGPA or a company's details. This ensures data remains current throughout the placement season.

*Account Deletion:* Admins can delete accounts for students who graduate or students who got placement or companies that no longer participate.

2. **Student Functions:**

   *Registration:* Students register by providing personal details like, name, roll number, email, phone**,** academic details like, CGPA, branch, year of study and uploading documents like, resume, certificates.

   *Profile Updates:* Students can update their profiles, including academic records, skills, and uploaded documents. The system ensures updates are reflected in real-time for job applications.

3. **Company Functions:**

   *Registration:* Companies register by providing details such as company name, industry, contact person, and email.

   *Profile Updates:* Companies can update job requirements, contact details, or company descriptions. This ensures accurate information is available to students and admins.

- **Placement Drive management:** To facilitate the institute and scheduling of placement drives, job postings, and interview rounds, ensuring efficient coordination between admins and companies.

1. **Admin Functions:**

   *Create Placement Drives:* Admins can schedule placement drives by specifying details such as company name, date, time, venue (physical or virtual), and interview rounds like, aptitude test, coding round, technical interview, HR interview.

   *Manage Drive Details:* Admins can update or cancel drives, notify affected users, and assign resources (e.g., rooms, interviewers).

   *Eligibility Configuration:* Admins define eligibility criteria for each drive (e.g., minimum CPI, specific branches like ICT or MnC or EVD). The system automatically filters eligible students.

2. **Company Functions:**

   *Post Job Openings:* Companies create job postings with details such as job title, description, post, location, eligibility criteria (e.g., CPI ≥ 7.0, no backlogs), and application deadlines.

   *Update Job Postings:* Companies can modify job details or extend deadlines. They can modify job posts i.e. which posts are vacant.

*View Drive Schedules:* Companies can view their scheduled drives and associated details (e.g., number of applicants, interview slots).

- **Job Application Management:** To enable students to apply for jobs, allow companies to shortlist candidates, and track the status of applications throughout the placement process.

1. **Student Functions:**

   *View Job Openings:* Students can view a list of job postings for which they are eligible, based on criteria like CPI, branch, or backlogs.

   *Track Application Status:* Students can view the status of their applications (e.g., Applied, Shortlisted, Interview Scheduled, Selected, Rejected).

Companies can access a list of applicants with their profiles (e.g., resume, CPI, skills).Companies can shortlist candidates for interview round via aptitude tests and coding round.

- **Reporting and Analytics:** To provide insights into placement performance, including placement rates, company participation, and student trends, to support decision-making.

*Placement Statistics:* Reports on placement success rates by department, batch, or company (e.g., percentage of students placed, average salary).

*Student Performance:* The system keeps track of each student's progress. For every student, it shows the companies they have applied to and the stages or rounds they have cleared in each company.

Admins can filter reports by criteria like year, department, or company and also we can do Comparative analysis (e.g., placement rates across years or departments).

- **Mentoring & Guidance Tracking**
  - **Mentor Mapping:** Assign faculty or alumni mentors to students.
  - **Session Tracking:** Store session dates, topics covered, and mentor feedback.
  - **Performance Insights:** Allow TPOs to identify students who may need additional guidance and resources.
- **Unique Selling Points (USP)**
  - Automated Eligibility Checks: Eliminates manual filtering, ensuring error-free student shortlisting.
  - End-to-End Drive Management: From job posting → application → shortlisting → interview scheduling → offer release, all handled in one platform.
  - Role-Specific Dashboards: Tailored views for students, recruiters, and admins.
  - Real-Time Data Synchronization: Ensures every update reflects instantly across the system.
  - Advanced Analytics: Placement trends, salary benchmarks, and student performance tracking help in strategic planning.

- **Unique Feature**

  The Placement Database Management System offers several unique features that make the placement process more efficient and transparent. It serves as a centralized platform for students, recruiters, and placement officers, providing secure access to each user. The system automates eligibility checks based on academic criteria. It manages placement drives, from job postings and student applications to shortlisting, interview scheduling, and final offer updates. Additionally, the system tracks the placement status of every student and provides detailed analytics and reports on placement statistics such as department-wise selections and package trends.

- **Benefits**

  - **For Students:** Easy access to opportunities, real-time updates, and transparency in the selection process.

  - **For Recruiters:** Streamlined hiring process, quick access to verified student data, and efficient scheduling.

  - **For Institutions:** Centralized control, reduced administrative workload, and valuable insights to improve placement rates.

- **Conclusion**

  The Placement Database Management System is a comprehensive solution designed to automate the placement process in academic institutions. By replacing traditional paper-based and spreadsheet-driven methods, it enhances efficiency, accuracy, and transparency in managing student, company, and placement-related data. Features like automated eligibility checks, placement drive management provide convenience to institutes or universities. The system not only simplifies the coordination between students, recruiters, and placement officers but also offers valuable insights through analytics and reporting tools. Overall, it ensures a more organized, reliable, and scalable approach to handling placements, ultimately benefiting students, institutions, and recruiting companies.

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## ❖ Background Reading/s

### ➢ Description of Each Reading Done & References

1. **How to write SRS - Software Requirements Specification Document**

   The video explains how to create an SRS (Software Requirements Specification), a formal document describing *what a system should do* and *how it should perform*.

- Purpose of SRS: Acts as a blueprint for developers avoids misunderstandings, and serves as a reference for validation and maintenance.

- Qualities of a Good SRS: Correct, complete, consistent, unambiguous, verifiable, and modifiable.

- Standard Structure (IEEE format):
  1. Introduction → purpose, scope, definitions, references.
  2. Overall Description → product perspective, functions, user characteristics, constraints, assumptions.
  3. Appendices → glossary, supporting information.

2. Placement Management System

The video demonstrates a Placement Management System project. It shows how to manage placement-related data (students, companies, and placement drives) using CRUD operations:

- **Create** → Add new student, company, or drive records.
- **Read** → View stored details from the database.
- **Update** → Modify existing records.
- **Delete** → Remove records.

It also explains the database schema, including tables for students, companies, and placements with relationships (primary and foreign keys). Overall, the video is a practical demonstration of how to build a functional Placement Management System with focus on database interactions.

3. Wikipedia

- A Software Requirements Specification (SRS) is a formal document that describes what a software system must do. The SRS clearly outlines both functional requirements (what the software should do) and non-functional requirements (how the software performs under constraints).

- An **SRS** is a comprehensive blueprint for any software project, outlining system behaviors, constraints, and usage scenarios. It creates a shared understanding among all stakeholders and significantly improves project alignment and execution. The IEEE and ISO/IEC/IEEE standards provide guidance on structure, quality, and best practices.

- Structure of document

  **Purpose** – Definitions, background, scope, and references.

**Overall Description** – Product perspective, system, user, hardware, software, and communication interfaces, constraints, assumptions, and dependencies.

4. scribd.com

- This Software Requirements Specification (SRS) defines the functions and constraints of a Placement Management System used by college placement cells to manage comprehensive student and company databases.

- It specifies capabilities like viewing, searching, creating, updating, and deleting student and company profiles, as well as sending placement-related notifications to students.

- The document is designed for Faculty members in the Training & Placement Office (TPO),Developers implementing the system,Students seeking placements,Company HR teams who can view and understand college placement data.

- The system manages both academic and personal student information, along with company profiles, including eligibility criteria and offered packages.

- It supports modifying or deleting records to keep data current.

- Search & Filtering: Find eligible students or companies based on defined parameters.

➢ **Combined Requirements Gathered from Background Readings**

The combined requirements represent a consolidation of insights gained from background readings, videos, and reference documents about Placement Management and Software Requirement Specification (SRS). They define what the system must do (functional requirements).

**1. Student Information Management**

Maintain detailed student profiles (Name, Roll No., Department, Contact details).Store academic records (CGPA, grades, year of study, backlogs).Store personal information (address, email, phone).Update, delete, or modify student records as needed. Provide eligibility filters (e.g., CGPA ≥ required, no backlogs).

**2. Company Information Management**

Maintain company profiles (Company name, domain, contact details).Store job role details (designation, package, eligibility criteria).Maintain details of placement drives (date, location, criteria).Update company requirements and placement data.

**3. Placement Drive Management**

Schedule company visits and notify eligible students.Track student participation in placement drives.Record outcomes of drives (selected, rejected, waiting list).Generate reports on placement statistics (per company, per student, per department).

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## ❖ Interviews

# Interview plan

**System :** Placement Database Management System

**Project reference :**

**Participants:** Kishan Rathod (Student)
Hardik Nakum (Student)

**Date:** DD/MM/YYYY **Time:**

**Duration:** 30-40 minutes **Venue:** as your choise

**Mode :** (Online/ in-person)

**Purpose Of Interview:**
To understand the placement process at DA-IICT in detail. To identify Strengths and weaknesses in current placement practices. To collect practical suggestions for improvement from the placement committee's perspective.

**Agenda:**
1. Placement Process
2. Placement Performance
3. Challenges with the current placement management process
4. suggestions and improvements.

From the role-play interviews with students and  placement committee members the following requirements were identified for the Placement Database Management System:

**Student Requirements:**

- Easy registration and profile creation (personal, academic, and skill details).
- Ability to upload resumes, certificates, and other supporting documents.
- Notifications about upcoming placement drives, deadlines, and company announcements.
- Dashboard to track application status, shortlisted companies, and offers.
- Self analytics and year wise analytics.

**Placement Cell Requirements:**

- Centralized database to manage all student and company records.

- Tools to schedule placement drives, interviews, and tests.
- Automated eligibility checking (based on CGPA, skills, or company criteria).
- Communication tools for sending updates, reminders, or announcements.

**Company Requirements:**

- Access to student profiles and resumes (with filters like CGPA, skills, or branch).
- Ability to post job openings and requirements.
- Smooth coordination for scheduling interviews and placement activities.

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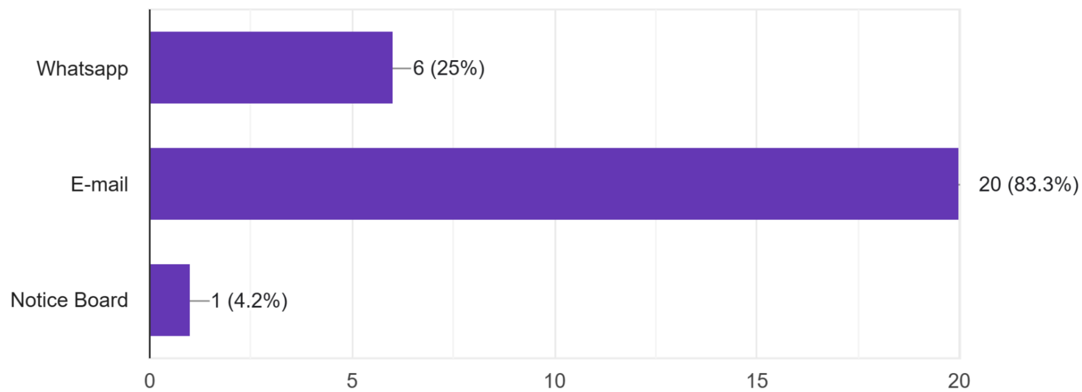
## ❖ Questionnaires

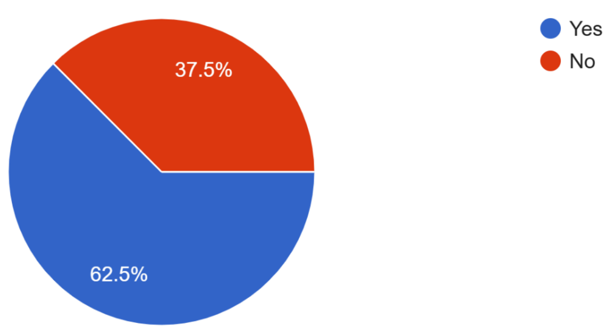
➔ From the responses, it was observed that most placement updates are received through email.

Question - 1 How do you currently get placement updates?
24 responses




➔ Many students face problems in the current placement system.

Question - 5 Do you face any problems in the current placement process?
24 responses



- ➔ I think mentoring provides students with guidance and support from faculty or alumni to help them prepare better for placements.
- ➔ Allows us to track our placement progress and compare skills with peers.

Here is the Link for our Google form.

Placement DB - Survey

- **Requirement Gathered**

  **Student Module** – Students need a platform to create/update profiles, upload resumes, enter academic details, check eligibility, and track their placement progress.

  **Company Module** – Placement officers require tools to register companies, post job details, announce drives, and maintain past recruitment records.

  **Placement Officer/Admin Module** – Officers need features to verify student data, manage postings, generate shortlists, schedule interviews, and prepare reports.

  **Notification System** – Students want instant alerts about new drives, application deadlines, interview schedules, and results via portal/SMS/Email.

  **Mentoring Support** – Students prefer mentor guidance (faculty/alumni) with session tracking and mentor feedback to improve preparation.

**Self-Analytics** – Students expect dashboards showing their placement statistics, skill-gap analysis, peer comparison, and progress tracking over time.

**Reports & Data Export** – Officers need detailed reports (student-wise, company-wise, year-wise) and export options for institutional use.

## ❖ Observations

Based on all interviews we observed our placement process structure.

DA-IICT operates a structured ecosystem featuring a Placement Cell, Student Placement Committee (SPC) ensuring a fair and transparent placement cycle.

The SPC, composed of faculty, staff, and student representatives, bridges students and recruiters, compiles essential student & company data, formulates placement policy, maintains communication, and handles certain logistics

**Company Outreach**: Invitations to recruiters are sent via emails, calls, or brochures; companies confirm by submitting a Job Announcement Form detailing roles, packages, visit dates, job descriptions, and recruitment processes

**Structured Ecosystem But Heavy Manual Overhead**

- The system involves SPC, faculty, and Placement Cell with a clear hierarchy, yet manual tasks like company outreach and policy management are prevalent.

**Dependence on Paper/Forms for Coordination**

- Recruiter engagement relies on Job Announcement Forms and emails—susceptible to inconsistencies and delays.

**Presentation-Centric Selection Flow**

- PPTs and interviews dominate the candidate experience, but personalized push and tracking are limited.

**Lack of Eligibility Automation & Data Access**

- Shortlisting appears largely manual; students rely on external channels to know status.

**Reporting & Statistics Not Fully Transparent**

- While macro-level stats are published, detailed breakdowns (e.g., by program/branch) are less visible, and alumni/students highlight gaps in transparency.

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## ❖ Fact Finding Chart

| Fact-Finding Method | Description | Sources/ Participants | Key Finding/ Requirements |
|---|---|---|---|
| Background Readings | Review of documents, videos, and websites to understand SRS structure and placement systems. This includes learning about SRS best practices and existing placement management tools. | - Video: "How to write SRS - Software Requirements Specification Document"<br>- Video: "Placement Management System"<br>- Wikipedia: Article on Software Requirements Specification (SRS)<br>- Scribd.com: SRS for Placement Management System<br><br>- Visit to DAU Placement website | - SRS structure: Introduction, Overall Description, Appendices<br>- Placement system functions: CRUD operations for students, companies, drives; database schema with relationships.<br>- Functional requirements: Manage student/company profiles, eligibility filters, notifications, search/filtering.<br><br>- Combined requirements: Student info management Company info, Placement drive management |

| **Fact-Finding Method** | **Description** | **Sources/ Participants** | **Key Finding/ Requirements** |
|---|---|---|---|
| Interviews | Role-play interviews to gather insights on the placement process, strengths, weaknesses, and suggestions. Focused on understanding user needs from different perspectives. | -Kishan Rathod (student)<br>-Hardik Nakum (Student)<br>-Ksitij Purohit (SPC member)<br>-Jay Limbasiya (SPC member) | - Student requirements: Easy registration, profile creation, document uploads, notifications, dashboard for status tracking, self/year-wise analytics.<br>- Placement Cell requirements: Centralized database, drive scheduling tools, automated eligibility checks, communication tools.<br><br>- Company requirements: Access to student profiles/resumes with filters, job posting, interview coordination. |

| Fact-Finding Method | Description | Sources/ Participants | Key Finding/ Requirements |
|---|---|---|---|
| Interview | Analysis of the current placement process at DA-IICT to identify structure, inefficiencies, and areas for improvement. | -Kishan Rathod (student)<br>-Hardik Nakum (Student)<br>-Rishid Unadkat (SPC Convener) | - Structured ecosystem with SPC bridging students/recruiters, but heavy manual overhead (e.g., emails, forms).<br>- Dependence on paper/forms leading to inconsistencies/ delays.<br>- Presentation-centric selection with limited personalization/tracking.<br>- Lack of automation in eligibility/shortlisting; incomplete transparency in reporting/stats.<br>- Need for automation to reduce manual tasks and improve data access/transparency. |

## ❖ List Of Requirements

| **Requirement** | Frequency |
|---|---|
| Maintain detailed student profiles (personal, academic, skills, documents) | 3 (Readings + Student Interview + Placement Cell) |
| Centralized database for student & company data | 3 (Placement Cell Interview + Observation + Readings) |
| Placement drive scheduling & management (tests, interviews, deadlines) | 3 (Placement Cell Interview + Company Interview + Readings) |
| Ability for students to register and update profiles | 2 (Student Interview + Readings) |
| Students can upload resumes, certificates, supporting documents | 2 (Student Interview + Readings) |
| Automated eligibility checking (CGPA, backlogs, skills, branch criteria) | 2 (Placement Cell Interview + Readings) |
| Students receive notifications/alerts (deadlines, drives, announcements) | 2 (Student Interview + Readings) |
| Students can track application status (applied, shortlisted, offers) | 2 (Student Interview + Readings) |
| Job posting system (companies add roles, criteria, salary, deadlines) | 2 (Company Interview + Readings) |
| Recruiters can filter/search student profiles (CGPA, skills, branch) | 2 (Company Interview + Readings) |

| | |
|---|---|
| Reporting & analytics (placement stats, offers, department-wise results) | 2 (Observation + Readings) |
| Replace manual/paper-based process with automated system | 2 (Observation + Description in SRS) |
| System shall provide notification alerts (deadlines, drives, interview schedules, status updates) | 2 (Student Interview + Questionnaire) |
| System shall provide mentoring support (faculty/alumni guidance, session tracking, feedback) | 2 (Student Interview + Questionnaire) |
| System shall provide self-analytics dashboards (progress tracking, peer comparison, skill-gap analysis) | 2 (Student Interview + Questionnaire) |

➔ After merging everything we have a clear requirement list.

1. The system shall maintain detailed student profiles including personal, academic, skills, and uploaded documents.
2. The system shall allow students to register, update, and manage profiles in real time.
3. The system shall allow students to upload resumes, certificates, and supporting documents.
4. The system shall provide a centralized database to manage student and company data.

5. The system shall include automated eligibility checking (based on CGPA, skills, branch, backlogs).

6. The system shall send notifications and alerts about placement drives, deadlines, and announcements.

7. The system shall allow students to track application status (applied, shortlisted, selected, rejected, offers).

8. The system shall support placement drive scheduling & management (tests, interviews, deadlines).

9. The system shall allow companies to post job openings with details (roles, criteria, salary, deadlines).

10. The system shall allow recruiters to filter/search student profiles (e.g., CGPA, skills, branch).

11. The system shall generate reports and analytics (placement statistics, offers, department-wise results).

12. The system shall replace manual/paper-based processes with automated workflows.
13. The system shall provide a notification module to send alerts about placement drives, approaching deadlines, interview schedules, and application status updates.
14. The system shall provide mentoring support by mapping students to faculty/alumni mentors, storing session details, and enabling coordinators to track mentoring effectiveness.
15. The system shall provide self-analytics dashboards for students to view their placement statistics, shortlisting ratio, peer comparison, and skill-gap analysis.

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## ❖ Classes and Characteristics

➢ **Students**

- Role: Register, update profiles, upload documents, apply for jobs, track application status.

- Characteristics: Large group, varied technical knowledge, need simple interface and mobile-friendly access.

➢ **Training and Placement Officers (TPOs)**

- Role: Manage placement cycles, approve company drives, configure eligibility, generate reports.

- Characteristics: Small, expert group with full system privileges. Need reliability, security, and detailed reporting.

➢ **Student Placement Committee (SPC)**

- Role: Assist TPOs in coordination, scheduling, and communication with students and companies.

- Characteristics: Student volunteers, technically aware, act as intermediaries.

➢ **Recruiters/Companies**

- Role: Post jobs/internships, set eligibility, view/filter applicants, schedule and conduct interviews.

- Characteristics: External users need secure access and minimal learning curve.

➢ **University Administration & Management**

- Role: Access analytics, placement statistics, and reports for accreditation and decision-making.

- Characteristics: Occasional users, require dashboards and read-only secure access.

➢ **Alumni (Mentoring Role)**

- Role: May provide career mentorship, referrals, or industry connections in the system.

- Characteristics: Occasional users, external to institute, limited contribution role.

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## ❖ Product Functions

1. **Student Data Management**
    - Store student records: personal info, academic history, CPI, skills, and uploaded resumes/certificates.
    - Support search and filter queries for academic/skill-based attributes.

2. **Company Data Management**
    - Maintain company details, job postings, eligibility criteria, and recruitment process info.
    - Store Job Announcement Forms (JAFs) digitally in structured tables.

3. **Application Handling**
    - Record student applications for each job role.
    - Track application status: applied, shortlisted, interview, offered, rejected.

4. **Eligibility Verification**
    - Use stored procedures/functions to automatically check student eligibility against company criteria (CPI, branch, skills, backlog).

5. **Placement Drive Scheduling**
    - Maintain schedules for tests, interviews, and group discussions.
    - Store timestamps, avoid conflicts, and log changes for audit.

6. **Reporting & Analytics (SQL Queries/Views)**
    - Generate institute-level statistics such as placed vs. unplaced students, company-wise offers, salary distributions, and program-wise trends.
    - Provide SQL views for coordinators and faculty to fetch summary data quickly.

7. **Role-Based Access Control (Database Level)**
    - Define roles in PostgreSQL:
        - Students: limited insert/update on their own profile, view jobs, see status.

        - Coordinators: full control on student/company/application tables.
        - Recruiters: read-only access to eligible student profiles.

8. **Mentoring & Guidance Tracking**
    - Map mentors (alumni) to students.
    - Store mentoring session details (date, topics, feedback, improvements).
    - Enable coordinators to monitor mentoring effectiveness through SQL reports.
9. **Self-Analytics (Student-Centric)**

- Provide personalized statistics for each student, such as:
    - Number of applications submitted.
    - Shortlisting ratio.
    - Comparison of CPI/skills vs. placed peers (aggregate from DB).
- Implement using SQL aggregate queries and materialized views.
- Helps students self-assess and improve their placement preparation.

## ❖ Privileges

| Function | Student | Recruiter | Placement Co-ordinator/ Admin |
|---|---|---|---|
| Student Data Management | Can insert & update only their own profile. | Read-only (only eligible student details). | Full access (insert, update, delete for all students). |
| Company Data Management | Read-only (can view job postings). | Can insert/update their own job postings (JAFs). | Full access (insert, update, delete for all companies). |
| Application Handling | Can insert (apply) and view/update own applications. | Can update status of applications (shortlist, reject, offer). | Full access (monitor all applications, override if needed). |
| Eligibility Verification | Read-only (see their eligibility status). | Read-only (view eligible candidates). | Full access (run eligibility functions, update criteria). |
| Placement Drive Scheduling | Read-only (view schedule). | Read-only (view schedule). | Full access (insert/update schedules, resolve conflicts). |
| Reporting & Analytics | Read-only (can view own self-analytics). | Read-only (company-wise analytics for their drives). | Full access (generate institute-level reports & export). |

| | | | |
|---|---|---|---|
| Role-Based Access Control | None (restricted). | None (restricted). | Full control (create/update roles & privileges). |
| Mentoring & Guidance | Insert/view own mentoring records. | None (not applicable). | Full access (assign mentors, view all sessions, reports). |
| Self-Analytics | Full access (view personal performance stats). | None (not applicable). | Full access (generate student analytics for monitoring). |

## ❖ Assumption

1. Each student has a unique enrollment number and each recruiter a unique company ID to ensure data consistency.

2. The placement process is seasonal (e.g., annual campus placement), and the system will be reset or archived after each season.

3. Student data (personal, academic, CPI, resumes, certificates) is entered correctly and verified by coordinators.

4. Company job postings (role, package, eligibility criteria) are assumed to be authentic and verified by placement officers before activation.

5. All documents (resumes, certificates, job descriptions) are uploaded in standard formats (PDF, DOCX) not exceeding fixed size limits.

6. Mentoring records will consist only of session metadata (mentor ID, mentee ID, date, feedback text) and not multimedia files.

7. Self-analytics uses existing database records only (applications, shortlists, offers) and will not require external data.

8. Students are expected to regularly update their profiles with latest CPI, skills, and certifications.

9. Recruiters will use the system to post jobs, filter candidates, and download resumes instead of external channels.

   Placement coordinators will manage scheduling of drives, interviews, and maintain reports directly in the system.

10. Notifications (deadlines, status updates, schedule changes) will be database-triggered and visible through the student dashboard.

11. Mentoring sessions are assumed to be offline/online meetings but only recorded in the database for reference.

12. Application status updates (shortlisted, rejected, offered) are assumed to be entered by recruiters or coordinators in a timely manner.

13. Data integrity will be maintained using constraints, triggers, and stored procedures in PostgreSQL.

14. Users will have stable internet/LAN access when using the system.

15. The system is assumed to handle a few thousand users and records at maximum load (typical for one placement season).

16. Heavy analytics (trend graphs, salary analysis) will be handled by SQL queries and views, not by external BI tools.

17. Notifications and reports will be generated in near real-time but not instantaneously (small delay acceptable).

## Summary

The **Placement Database Management System (PDMS)** is designed to automate and streamline the entire campus placement process. It replaces traditional paper-based and spreadsheet-driven methods with a centralized database solution built on PostgreSQL. The system reduces manual workload, eliminates communication gaps, and ensures transparency for all stakeholders, including students, recruiters, Training & Placement Officers (TPOs), and university management.

The PDMS maintains detailed student and company profiles, manages placement drives, and enables secure role-based access. Students can register, update profiles, upload resumes, apply to opportunities, and track application status in real time. Companies can post job openings, define eligibility, shortlist candidates, and manage interviews, while TPOs oversee scheduling, eligibility verification, and reporting.

Key features include:

**Student Data Management**

- Maintain detailed student profiles (personal, academic, skills, documents).
- Students can register, update profiles, and upload resumes/certificates in real time.

**Company Data Management**

- Store company profiles, job postings, eligibility criteria, and recruitment details.
- Companies can add, update, and manage job announcement forms (JAFs).

**Placement Drive Scheduling & Management**

- Schedule aptitude tests, coding rounds, interviews, and HR sessions.
- Avoid scheduling conflicts and notify students automatically.

**Application Handling**

- Students can apply to eligible jobs and track application status (applied, shortlisted, selected, rejected, offers).
- Recruiters can shortlist candidates and update application outcomes.

**Automated Eligibility Verification**

- System automatically checks student eligibility based on CPI, branch, backlogs, and skill requirements.

**Reporting & Analytics**

- Generate placement statistics (department-wise, year-wise, company-wise).
- Comparative analysis across batches, average/median salary reports, and student performance tracking.

**System Notifications**

- Real-time alerts for placement drives, approaching deadlines, interview schedules, and application status changes.
- Role-based notifications for students, recruiters, and coordinators.

**Mentoring & Guidance Support**

- Map students to faculty/alumni mentors.
- Store mentoring session details (date, topics, feedback).
- Coordinators can monitor mentoring effectiveness with reports.

**Self-Analytics (Student-Centric)**

- Dashboards for students showing number of applications, shortlisting ratio, and progress over time.
- Peer comparison (CPI/skills vs. placed peers) and skill-gap analysis for self-improvement.